

MAT1600 Lecture Notes

Arky!! :3c

'26 Fall Semester

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Day 1: Measure Theory (Sep. 9, 2026)

The syllabus is on the Quercus page, with contact information for professor and TA available. Office hours will be held on 2–3pm on Wednesdays. They’re going to be in the Huron lounge. The grading scheme is as follows,

- (i) (15%) 6 assignments, roughly due every two weeks, graded for completeness. Must be handwritten.¹
- (ii) (20% × 2) Two midterms, both in-class on Mondays (Oct. 19, Nov. 16)
- (iii) (45%) Final.

On the graduate side of things, most of the grades will be an “A–” (As and A+s are mostly for department recognition). A *B-range* is an indication that the instructor would like you to spend more time with the material (i.e., a qualifying exam). The midterms would be... ideally be able to distinguish stronger students from weaker students. “On the higher end, everything looks identical for graduate students and undergraduate students... on the lower end, the undergraduate curve runs... lower.”

The first three classes will be mostly measure theoretic. Things like Carathéodory’s extension theorem, construction of Lebesgue measure, etc... (things covered in MAT457) will be stated, but not proved.

We now begin with “formalizing probability”.

Example 1.1 (Coinflips). Let’s say we flip 2 coins; there are 4 possibilities: you can get (with *H* for heads and *T* for tails) *HH*, *HT*, *TH*, *TT*. Assuming we’re flipping fair coins, we have that the probabilities for each outcome is equal,

$$\mathbb{P}(HH) = \mathbb{P}(HT) = \mathbb{P}(TH) = \mathbb{P}(TT) = \frac{1}{4}.$$

We may, for example, study the probability to flip at least one head. Clearly,

$$\mathbb{P}(\text{at least one } H) = \mathbb{P}(\{HH, HT, TH\}) = \mathbb{P}(HH) + \mathbb{P}(HT) + \mathbb{P}(TH) = \frac{3}{4}.$$

We may describe $\{HH, HT, TH\}$ as an *event*, i.e., a set of outcomes. We can describe the set of all outcomes, as described earlier as the four possible coinflip results, as Ω , our probability space. In this manner, events can be conceptualized as subsets of Ω .

Example 1.2 (Infinitely many coinflips). Now, let’s flip infinitely many (fair) coins. The outcomes are now sequences of *H* and *T*. We have the following observations,

$$\mathbb{P}(\text{at least one } H) = 1, \quad \mathbb{P}(TTTT \dots) = 0.$$

We also have that

$$\mathbb{P}(\text{all tails}) \leq \mathbb{P}(\text{first } k \text{ coins are tails}) = \frac{1}{2^k}.$$

Since the above is true for all k , we can take $k \rightarrow \infty$ to get that the probability of flipping all tails is indeed 0.

Per the two examples above, it follows that probability spaces need not be very nice, which is why we’re going to use measure theory. As in the example for infinitely many coinflips, we have a few problems. While outcomes have probability 0, specific events do not. We cannot (nor do we want to) study *all events*; we will only study a “nice” subset of them.

¹i can complete the assignments in a *non-handwritten* way in a short amount of time... i’ll let you think about what i mean...

Definition 1.3 (σ -algebra). Let Ω be a set. A σ -algebra $\mathcal{A}$ on Ω is a nonempty collection of subsets of Ω such that

- (i) $A \in \mathcal{A}$ implies that $A^c \in \mathcal{A}$,
- (ii) For countable collections $(A_n : n \in \mathbb{N})$ in $\mathcal{A}$, we have that

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

Note that it is redundant to state that $\emptyset \in \mathcal{A}$; since $\mathcal{A}$ was already supposed to be nonempty, we could simply pick $A \in \mathcal{A}$ and see that $\emptyset = (A \cup A^c)^c \in \mathcal{A}$. A pair $(\Omega, \mathcal{A})$ is called a *measurable space*.

Definition 1.4 (Probability measure). A probability measure on a measurable space $(\Omega, \mathcal{A})$ is a function $\mathbb{P} : \mathcal{A} \rightarrow [0, 1]$ such that²

- (i) $\mathbb{P}(\Omega) = 1$,
- (ii) $\mathbb{P}$ is countably additive, i.e., if $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$ is a disjoint collection of sets, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

A triple $(\Omega, \mathcal{A}, \mathbb{P})$ is a *probability space*.

Exercise 1.5. If $(\mathcal{A}_i : i \in I)$ is any collection of σ -algebras on Ω , then $\bigcap_{i \in I} \mathcal{A}_i$ is a σ -algebra.

Exercise 1.6. For any collection of sets $\mathcal{B} \subset 2^\Omega$, there is a *smallest* σ -algebra $\sigma(\mathcal{B})$ containing $\mathcal{B}$.

We will denote $\sigma(\mathcal{B})$ to be the intersection of all σ -algebras containing $\mathcal{B}$.

Fact 1.7. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space.

- (i) (Monotonicity). If $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- (ii) (Subadditivity / Union bound). For $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$, we have that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

- (iii) (Continuity). Let $(A_n : n \in \mathbb{N}) \subset \mathcal{A}$. If

$$A_1 \subset A_2 \subset \dots$$

and $A = \bigcup_{n=1}^{\infty} A_n$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$. Similarly, if

$$A_1 \supset A_2 \supset \dots$$

and $A = \bigcap_{n=1}^{\infty} A_n$, then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$.

As a quick proof of monotonicity, we simply have $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) \geq \mathbb{P}(A)$.

²“something has to happen” – Duncan; “nothing ever happens” – Daniel

We then took a 1 hour break, and Duncan came back to talk about a “preview” of this course; we will be looking at things like (weighted) random walks (i.e., the four cardinal directions). We will also look at the law of large numbers, the central limit theorem, and so on.

We now continue talking about 1.7.

Proof. To see continuity from below, observe that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n),$$

per the chain of inclusion. We can also write it as

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} (A_n \setminus A_{n-1})\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n \setminus A_{n-1}) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{P}(A_n \setminus A_{n-1}) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N). \quad \square$$

The other two parts of 1.7 are left as an exercise.

We now look at examples of probability measures.

- (i) Let Ω be finite or countable, and take $\mathcal{A} = 2^{\Omega}$, with a function $p: \Omega \rightarrow [0, 1]$, where $\sum_{\omega \in \Omega} p(\omega) = 1$. Then

$$\mathbb{P}(A) = \sum_{\omega \in A} p(\omega)$$

defines a probability measure.

- (ii) (Measures on $\mathbb{R}$). Let $\mathcal{B}(\mathbb{R})$ be the Borel σ -algebra on $\mathbb{R}$ (i.e., the smallest σ -algebra containing all open sets).

Theorem 1.8. There is a probability measure $\mathbb{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ satisfying

$$\mathbb{P}((-\infty, x]) = F(x)$$

for some function $F: \mathbb{R} \rightarrow [0, 1]$ if and only if (i) F is non-decreasing and right-continuous, (ii) $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow +\infty$. We call F the *CDF*, or *cumulative distribution function* of $\mathbb{P}$.

Proof. For the easy (only if) direction, monotonicity of $\mathbb{P}$ yields a non-decreasing F . Continuity from above yields right-continuity; specifically, if $x_n \downarrow x$, then

$$\bigcap_{n=1}^{\infty} (-\infty, x_n] = (-\infty, x],$$

so $\mathbb{P}(-\infty, x] = \lim_{n \rightarrow \infty} \mathbb{P}(-\infty, x_n]$. For the last parts, $F(x) \rightarrow 0$ (as $x \rightarrow -\infty$) uses continuity from above, and $F(x) \rightarrow 1$ (as $x \rightarrow \infty$) uses continuity from below.

The hard (if) direction comes from the Carathéodory extension theorem (existence of $\mathbb{P}$), and the uniqueness was not mentioned. $\square$

- (iii) Let $\mathbb{Q}$ be a σ -finite measure on $(\Omega, \mathcal{A})$. Let $f \in \Omega \rightarrow [0, \infty)$ be measurable such that

$$\int f(\omega) d\mathbb{Q}(\omega) = 1.$$

Then $\mathbb{P}(A) = \int f(\omega) \mathbf{1}(\omega \in A) d\mathbb{Q}(\omega)$ is a probability measure on Ω . f is called the *density*, or the *Radon–Nikodym derivative*. As an example,

$$\mathbb{P}(A) = \int \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \mathbf{1}(x \in A) dx$$

is the standard Gaussian measure on $\mathbb{R}$ (with its associated distribution). For another example,

$$f(x) = \frac{e^{-\lambda x}}{\lambda} \mathbf{1}(x \geq 0)$$

is the exponential distribution with parameter $\lambda \geq 0$. For a third example,

$$f(x) = \frac{\mathbf{1}(x \in [a, b])}{b - a}$$

is the uniform distribution on $[a, b]$.

Day 2: (Sep. 14, 2026)

Office hours will be held by Nitit (the TA) on Mondays, from 5 to 6pm.

Today, we will talk about Dynkin's π - λ theorem (cf. p.5 of Panchenko's notes).

Definition 2.1. A π -system is a collection of sets $\mathcal{P}$ which is closed under finite intersection, i.e.,

$$A, B \in \mathcal{P} \implies A \cap B \in \mathcal{P}.$$

A λ -system is a nonempty collection of sets $\mathcal{L}$ closed under complements and countable disjoint unions, i.e.,

$$A \in \mathcal{L} \implies A^c \in \mathcal{L},$$

and if $(A_i : i \in \mathbb{N})$ is a countable collection of disjoint sets in $\mathcal{L}$, then their union $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{L}$.

Theorem 2.2 (Dynkin). If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system such that $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Recall that we denote $\sigma(\mathcal{P})$ to be the σ -algebra generated by $\mathcal{P}$, i.e., the smallest σ -algebra containing $\mathcal{P}$ (obtained by taking the intersection over all such σ -algebras).

As an example of how one might use 2.2, let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on a measurable space $(\Omega, \mathcal{S})$. Suppose, for some π -system $\mathcal{P}$, that we have $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \mathcal{P}$. Then $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \sigma(\mathcal{P})$.

Proof. Let $\mathcal{A} = \{A \in \mathcal{S} : \mathbb{P}(A) = \mathbb{Q}(A)\}$. $\mathcal{A}$ is a λ -system, since

$$\begin{aligned} \mathbb{P}(A) = \mathbb{Q}(A) &\implies \mathbb{P}(A^c) = 1 - \mathbb{P}(A) \\ &= 1 - \mathbb{Q}(A) = \mathbb{Q}(A^c), \end{aligned}$$

whence it is closed under complements; in particular, it is also closed under countable disjoint unions by countable additivity of $\mathbb{P}$ and $\mathbb{Q}$. Since $\mathcal{P} \subset \mathcal{A}$, it follows that 2.2 would yield $\sigma(\mathcal{P}) \subset \mathcal{A}$ as desired. $\square$

We now prove Dynkin's π - λ theorem.

Proof (of 2.2). First, we want to show that " $\pi + \lambda = \sigma$ ", i.e., if $\mathcal{A}$ is both a π -system and a λ -system, then it is a σ -algebra. It suffices to check that $\mathcal{A}$ is closed under countable unions: if $(A_i : i \in \mathbb{N})$ is in $\mathcal{A}$, then we may write

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} A_i \setminus \left(\bigcup_{j=1}^{i-1} A_j \right) = \bigcup_{i=1}^{\infty} \left(A_i \cap \bigcap_{j=1}^{i-1} A_j^c \right)$$

to realize the union of $(A_i : i \in \mathbb{N})$ as a countable disjoint union, whence we are done per definition of λ -systems.

Thus, it is enough to show that $\lambda(\mathcal{P})$ (which we will hereby denote as the λ -system generated by $\mathcal{P}$) is a π -system, since from there, we can finish by arguing:

- $\lambda(\mathcal{P})$ exists, since being a λ -system is closed under arbitrary intersections.
- Then $\mathcal{P} \subset \lambda(\mathcal{P}) \subset \mathcal{L}$.
- From earlier, it would follow that $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.

For a fixed set A , let $\mathcal{G}_A = \{B \in \Omega : B \cap A \in \lambda(\mathcal{P})\}$. We will show that if $A \in \lambda(\mathcal{P})$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_A$, in three steps as below:

(i) If $A \in \lambda(\mathcal{P})$, then $\mathcal{G}_A$ is a λ -system. Let $(B_i : i \in \mathbb{N})$ be disjoint sets in $\mathcal{G}_A$. Then

$$\left(\bigcup_{i=1}^{\infty} B_i \right) \cap A = \bigcup_{i=1}^{\infty} (B_i \cap A),$$

for which each of $B_i \cap A$ is in $\lambda(\mathcal{P})$, since $B_i \in \mathcal{G}_A$. For complements, let $B \in \mathcal{G}_A$. Then

$$B^c \cap A = ((B \cap A) \cup A^c)^c \in \lambda(\mathcal{P}),$$

so $B^c \in \mathcal{G}_A$ as well. Thus, $\mathcal{G}_A$ is a λ -system.

- (ii) Now, if $A \in \mathcal{P}$, then $\lambda(\mathcal{P}) \in \mathcal{G}_A$. If we can show that $\mathcal{P} \subset \mathcal{G}_A$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_A$, since $\mathcal{G}_A$ is a λ -system by (i). Why is $\mathcal{P} \subset \mathcal{G}_A$? If $B \in \mathcal{P}$, then $B \cap A \in \mathcal{P}$, and so $B \cap A \in \lambda(\mathcal{P})$ implies $B \in \mathcal{G}_A$.
- (iii) If $B \in \lambda(\mathcal{P})$, then $\lambda(\mathcal{P}) \subset \mathcal{G}_B$. By (ii), $\mathcal{P} \subset \mathcal{G}_B$, and by (i), $\mathcal{G}_B$ is a λ -system. Since $\lambda(\mathcal{P})$ is the smallest λ -system (given by the intersection of all λ -systems containing $\mathcal{P}$), we have that $\lambda(\mathcal{P}) \subset \mathcal{G}_B$. Note that this concludes the property that $\lambda(\mathcal{P})$ is a π -system, per definition of $\mathcal{G}_B$. $\square$

For a more concise proof, see pg. 6 in Panchenko.